TCLNET: LEARNING TO LOCATE TYPHOON CENTER USING DEEP NEURAL NETWORK

Chao Tan

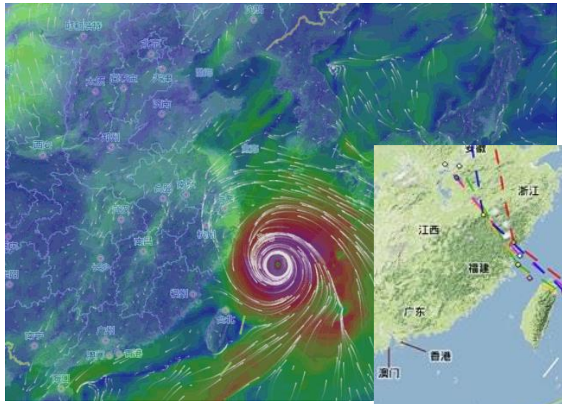
School of Computer Science and Engineering

Chongqing University of Technology

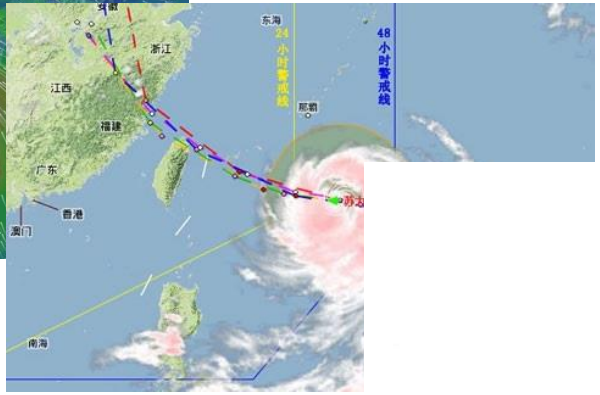
istvartan@outlook.com

Task and Applications

Typhoon Center Location: Determine the position of typhoon eye or spin center by a single frame infrared satellite cloud imagery.



a. typhoon intensity estimation



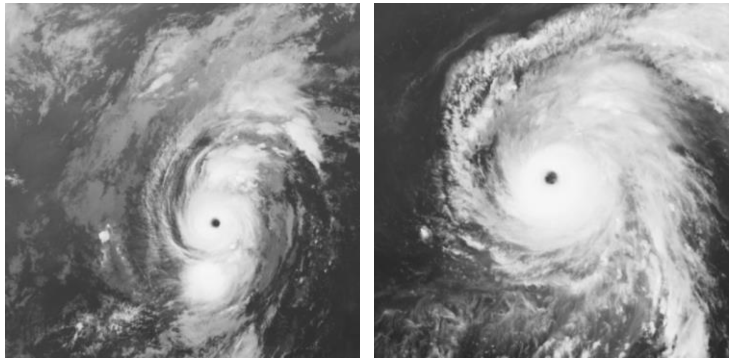
b. typhoon path forecasting

TC PLUS

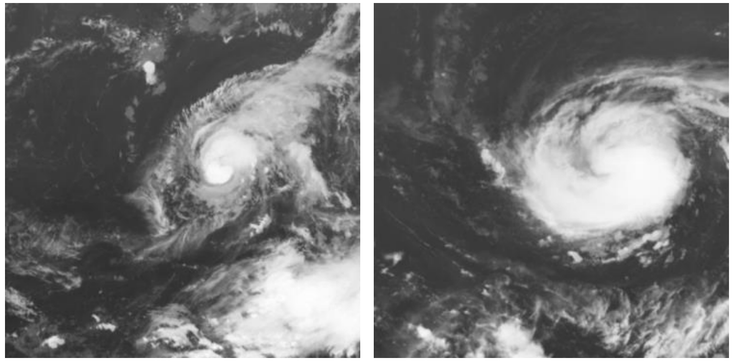
c. typhoon center tracking

Typhoon Center Location Dataset

TCLD: The first large-scale **T**yphoon **C**enter **L**ocation **D**ataset for deep learning research



a. eyed typhoon samples in TCLD



b. non-eyed typhoon samples in TCLD

Evaluation Metrics:

$$\text{MLE} = \frac{\sum_{i=1}^{n}\sqrt{(x^i - u^i)^2 + (y^i - v^i)^2}}{n}$$

MLE: Mean Location Error
n: number of testing samples.
(x,y): predicted typhoon center coordinate.
(u,v): label coordinate.

Human Benchmark (MLE):

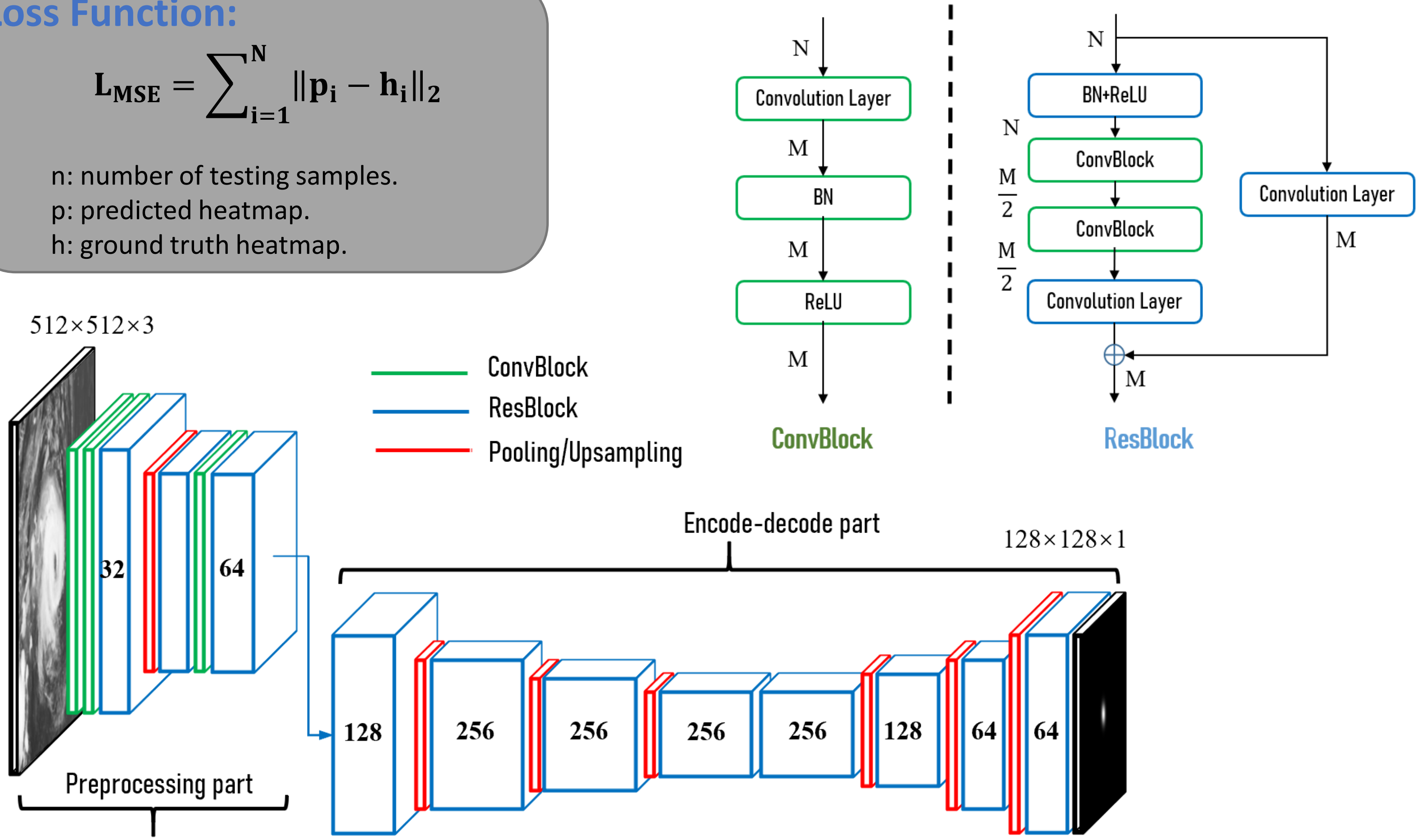
EYED	NON-EYED	AVERAGE
1.6800	6.5465	3.3886

Lightweight Network Structure (TCLNet)

Loss Function:

$$L_{MSE} = \sum_{i=1}^{N} \|p_i - h_i\|_2$$

n: number of testing samples.
p: predicted heatmap.
h: ground truth heatmap.



The overall network structure of TCLNet. It consists of a preprocessing part and encode-decode part stacked by **ConvBlocks**, **ResBlocks** and **Pooling/Upsampling layers**. The number of convolution kernels is marked on each ResBlock. TCLNet takes an infrared cloud image of 512×512 size as input and outputs a 128×128 single-channel heatmap.

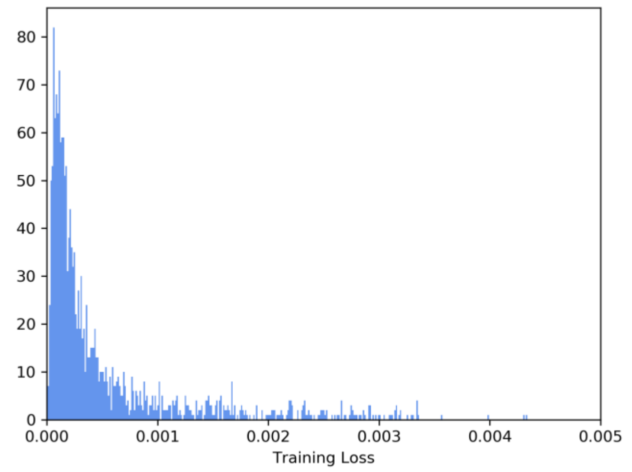
Experiments & Result Comparison

MODEL	MLE-AVERAGE	MLE-EYED	MLE-NON_EYED	PARAMETERS
ResNet50	51.299±1.0469	46.187±0.9739	60.747±1.4753	23.512M
SimpleBaseline-ResNet50	6.7906±0.1275	4.5178±0.1607	10.991±0.3168	33.996M
SimpleBaseline-ResNet50-up	6.2601±0.0588	5.1693±0.1968	8.2760±0.2562	29.409M
FPN-ResNet50	5.5177±0.0440	3.7764±0.0572	8.7358±0.1329	24.965M
CPN-ResNet50	5.3182±0.0848	3.3225±0.1245	9.0067±0.1301	27.222M
CPN-ResNet101	5.8058±0.2934	3.7922±0.0823	9.5274±0.7770	46.215M
Hourglass Network (×3)	5.2757±0.1490	3.5781±0.2143	8.4134±0.2130	14.998M
TCLNet (ours)	**4.5137**±0.0846	**2.8934**±0.0620	**7.5083**±0.1684	**1.0959M**

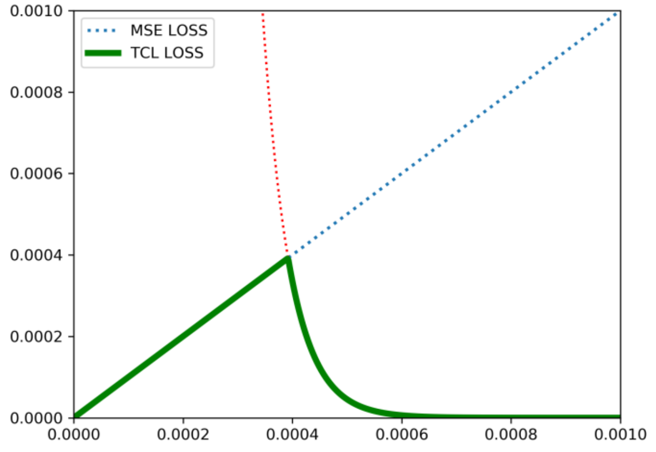
References:

He K, Zhang X, Ren S, et al. Deep residual learning for image recognition. CVPR 2016.
Xiao B, Wu H, Wei Y. Simple baselines for human pose estimation and tracking. ECCV 2018.
Lin T Y, Dollár P, Girshick R, et al. Feature pyramid networks for object detection. CVPR 2017.
Chen Y, Wang Z, Peng Y, et al. Cascaded pyramid network for multi-person pose estimation. CVPR 2018.
Newell A, Yang K, Deng J. Stacked hourglass networks for human pose estimation. ECCV 2016.
Yang X, Zhan Z, Shen J. A Deep Learning Based Method for Typhoon Recognition and Typhoon Center Location. IGARSS 2019.

TCL+ Loss Function



Loss distribution of different samples during training.



The function image of proposed TCL+ loss (solid green line).

TCL+ Loss Function:

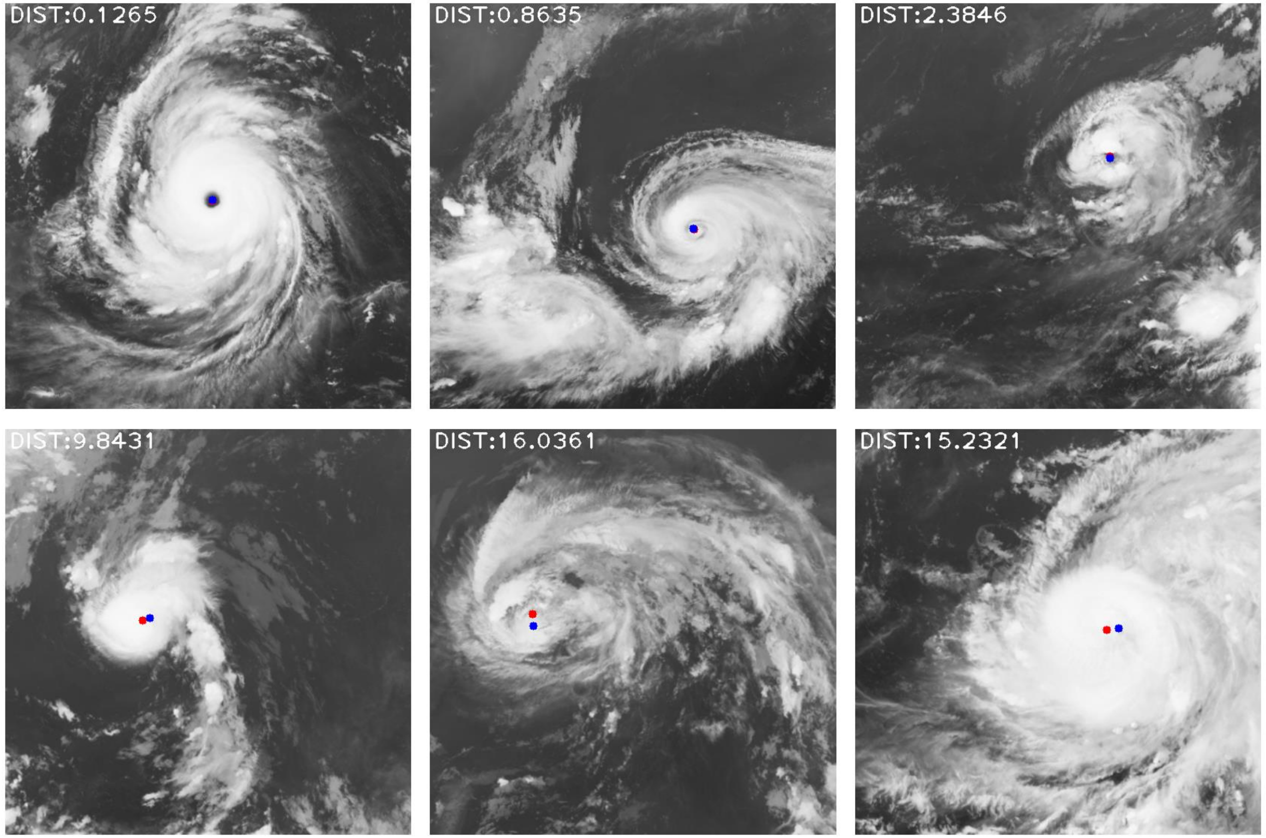
$$\mathbf{L_{TCL+}} = \mathbf{min}(\mathbf{L_{MSE}}, \mathbf{exp}(-\mathbf{2} \times \mathbf{10^4} \times \mathbf{L_{MSE}}))$$

$$\mathbf{L_{MSE}} = \sum_{\mathbf{i=1}}^{\mathbf{N}} \|\mathbf{p_i} - \mathbf{h_i}\|_{\mathbf{2}}$$

Experimental Results:

MODEL	MLE-AVERAGE	MLE-EYED	MLE-NON_EYED
Human Error	3.3886	1.6800	6.5465
TCLNet	4.5137±0.0846	2.8934±0.0620	7.5083±0.1684
TCLNet (TCL+)	**4.4389**±0.0287	**2.8462**±0.0305	**7.3825**±0.1170

Experimental Visualization



Top: three eyed typhoon samples with small errors. **Bottom**: three non-eyed typhoon samples with large errors. MLE is shown in the upper-left of each image. Blue dots represent the typhoon center labels, and red dots represent the predicted centers.

Further Studies on TCLNets

NO.	HOURGLASS	SCALES	SKIP	MLE-AVERAGE	MLE-EYED	MLE-NON_EYED	PARAMETERS
1	3	5	√	5.2757±0.1490	3.5781±0.2143	8.4134±0.2130	14.998
2	2	5	√	5.2838±0.0873	3.4844±0.1201	8.4369±0.3202	10.099
3	1	5	√	5.3050±0.0720	3.3696±0.0914	9.0366±0.2726	5.2012
4	1	4	√	5.2895±0.0526	3.3031±0.1623	8.9597±0.3256	4.3602
5	1	3	√	5.2547±0.0222	3.4891±0.0524	8.5178±0.1203	3.5192
6	1	2	√	5.3316±0.0854	3.3937±0.1986	8.9133±0.3238	2.6783
7	1	1	√	6.3172±0.2273	3.5399±0.1018	11.450±0.6130	**1.8373**
8	1	3	×	**4.9839**±0.0281	**3.2481**±0.0962	**8.1922**±0.1617	2.1563

Experimental results based on different network structure settings. **NO**. denotes the number of experiment. **HOURGLASS** means the number of hourglass module. **SKIP** means using skip connection.

NO.	MODELS	MULTI	MLE-AVERAGE	MLE-EYED	MLE-NON_EYED	PARAMETERS
1	256×2→256×2→256×2	×	5.2099±0.0870	3.4524±0.2104	8.4581±0.2101	3.6580
2	128×2→128×2→128×2	×	5.0860±0.0897	3.2555±0.1542	8.4693±0.1369	0.9238
3	256→256→256	×	4.9839±0.0281	3.2481±0.0962	8.1922±0.1617	2.1563
4	128→128→128	×	4.7929±0.0318	3.4635±0.0658	7.2500±0.0879	0.5456
5	64→64→64	×	4.9639±0.0726	3.3258±0.0814	7.9920±0.3273	**0.1397**
6	64→128→128	×	4.6698±0.0602	3.1351±0.0845	7.5065±0.1424	0.3962
7	64→128→128	√	4.7372±0.0323	3.0379±0.0464	7.8778±0.0460	0.6089
8	64→128→256 (**TCLNet**)	×	**4.5137**±0.0846	**2.8934**±0.0620	**7.5083**±0.1684	1.0959

Experimental results based on different number of convolution filters settings. **MULTI** denotes use multi-scale deep supervision or not.

Further Studies on TCLNets

GT Heatmap Generation:

$$\mathbf{H}(\mathbf{x},\mathbf{y}) = \mathbf{exp}\left(\frac{(\mathbf{x}-\boldsymbol{\alpha}\times\mathbf{u})^2+(\mathbf{y}-\boldsymbol{\alpha}\times\mathbf{v})^2}{-2\boldsymbol{\sigma}^2}\right)$$

H(x,y): value of ground-truth heatmap at (x,y).
(u,v): typhoon center coordinate.
α: scaling factor, defaults to 4.
σ: standard deviation, defaults to 15.

Experimental Results:

STD (σ)	MLE-AVERAGE	MLE-EYED	MLE-NON_EYED
05	5.1253±0.0741	3.4311±0.1437	8.2564±0.3290
10	4.6345±0.0417	3.0714±0.1158	7.5235±0.1912
15	**4.5137**±0.0846	**2.8934**±0.0620	**7.5083**±0.1684
20	4.7810±0.1145	3.0845±0.0616	7.9165±0.3020
25	4.9612±0.0391	3.2148±0.0640	8.1890±0.1735
30	5.7168±0.1376	4.2104±0.2082	8.5009±0.2371

Experimental results of different standard deviations used to generate heatmap labels. **STD** means the standard deviations.

THANKS

- source code
- daatset
- pre-trained model

- project page